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Advancing Predictive ADMET Modeling Through Community-Driven Science: The ExpansionRx-OpenADMET Blind Challenge



Community Article

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We're excited to announce that **the ExpansionRx-OpenADMET Blind Challenge is now live on Hugging Face!**

This community challenge aims to advance benchmarking of predictive models for Absorption, Distribution, Metabolism, Excretion, and Toxicology (ADMET) properties by providing a high-quality experimental dataset that participants can use to train and evaluate their models.

The Challenge

Small molecules continue to be the foundation of modern drug discovery, representing nearly 75% of FDA drug approvals in the last decade. Their ease of synthesis and tunable properties make them highly versatile therapeutic agents. However, predicting how these molecules behave in the body (including how long

they persist, where they go, and whether they interact with other targets), is extremely challenging. Collectively, these **ADMET** properties sit at the heart of the assay cascade and can make or break preclinical candidate molecules.

OpenADMET is an open-science initiative that aims to tackle these challenges by integrating structural biology, high-throughput experimentation, and computational modeling to improve ADMET prediction (read more about their strategy [here](#)). A key part of these efforts is organizing blind challenges to benchmark the current state of predictive modeling on real, high-quality datasets.

In partnership with ExpansionRx, this challenge invites participants to solve realistic ADMET prediction problems that ExpansionRx encountered during lead optimization, by predicting the properties of late-stage molecules based on earlier-stage data from the same campaigns, across nine endpoints.

A high-quality dataset of ADMET endpoints

Expansion Therapeutics recently prosecuted several drug discovery programs targeting RNA-mediated diseases, including Myotonic Dystrophy (DM1), Amyotrophic Lateral Sclerosis (ALS), and Dementia. During optimization, they generated a wealth of high-quality ADMET data off-targets and properties of interest. Now, they've made the bold and generous decision to open-source their high-quality ADMET dataset for public use and benefit.

The dataset includes over 7,000 small molecules measured across multiple ADMET assays, and was divided into:

- A **training set**, which will be available to all participants during the challenge
- A **blinded test set**, which will be used to score final predictions after submission.

Participants will train models on the training data, submit predictions for the molecules in the blinded test set, and have their models evaluated on the unseen

datapoints. ADMET Endpoints to predict There are a total of **nine ADMET endpoints** that participant is tasked with predicting:

- **LogD:** Measures compound's lipophilicity at a specific pH, which helps understand the balance of aqueous solubility with membrane permeability.
- **Kinetic Solubility KSOL:** Quantifies how much a compound can be dissolved under non-equilibrium conditions (in μM units).
- **Human Liver Microsomal (HLM) CL_{int}:** Helps predict in vivo liver metabolism and clearance (in mL/min/kg).
- **Mouse Liver Microsomal (MLM) stability:** Analogous to HLM, studying MLM can provide a more comprehensive understanding of a compound's metabolic profile and how a compound might behave in multi-species preclinical development. (in mL/min/kg).
- **Caco-2 Papp A>B:** Measures the rate of flux of a compound across polarized Caco-2 cell monolayers from the apical (intestinal lumen-facing side) to basolateral (blood-facing side), effectively mimicking the absorption of a drug across the intestinal wall (measured in 10^{-6} cm/s).
- **Caco-2 Efflux Ratio:** Models intestinal absorption by measuring flux across polarized Caco-2 cell monolayers.
- **Mouse Plasma Protein Binding (MPPB):** Determines the concentration of free drug in plasma (as % Unbound).
- **Mouse Brain Protein Binding (MBPB):** Measures the fraction of drug not bound to proteins within brain tissue (as % Unbound):
- **Mouse Gastrocnemius Muscle Binding (MGMB):** Reflects the amount of drug free to act within skeletal muscle tissue, which is important for drugs targeting peripheral or muscular conditions (as % Unbound)

How to get started

The challenge is run from a Hugging Face Space. Explore the challenge space [here](https://huggingface.co/blog/hugging-science/the-expansionrx-openadmet-blind-challenge).

Feel free to ask any questions on our community [Discord](#). The organizers and other participants will be able to assist you.

[🔗](#) Run through the tutorial:

We have a set of tutorials available for you to get acquainted with the challenge [here](#).

We recommend that all participants run through the tutorial to fully understand the endpoints and submission process.

[🔗](#) Access the training set:

Click [here](#) to see the dataset on the Hugging Face Hub, or you can easily download the clean version of the training dataset using the Hugging Face Datasets library:

```
from datasets import load_dataset

ds = load_dataset("openadmet/openadmet-expansionrx-challenge-train-dataset")
df = ds["train"].to_pandas()
```

And to download the raw version of the training dataset that contains out of bounds measurements.

```
from datasets import load_dataset

ds = load_dataset("openadmet/openadmet-expansionrx-challenge-train-dataset")
df = ds["train"].to_pandas()
```

[🔗](#) Access the blinded test set:

See the dataset on the Hub [here](#).

The test set contains only the SMILES strings for you to perform inference on before submitting to the challenge space.

Similarly to the training set you can access the blinded test data using the dataset library:

```
from datasets import load_dataset
ds = load_dataset("openadmet/openadmet-expansionrx-challenge-test-data")
df = ds["test"].to_pandas()
```

By taking part in this blind challenge, you'll contribute to open, reproducible ADMET modeling and help benchmark the next generation of predictive models for drug discovery!

[🔗](#) Questions or Ideas?

OpenADMET would love to hear from you whether you're:

- Interested in participating
- Have ideas for future challenges
- Want to contribute data

Join the OpenADMET [Discord](#) or email openadmet@omsf.io.

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